

# Alyssa Johnson

**Quantitative Researcher** | Physics PhD candidate developing ML models for signal detection, forecasting, and alpha-driven financial modeling.

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## SKILLS

**Quantitative Finance:** Options Pricing & Greeks, Volatility Modeling, Statistical Inference, Bayesian Modeling, Stochastic Processes, Signal Processing, A/B Testing, Anomaly Detection, Bloomberg Terminal & OpenBB

**ML / Stats:** Time Series Forecasting, Tree-Based & Ensemble Models (XGBoost, LightGBM), Deep Learning, NLP/LLMs, RAG Systems, Vector Embeddings, Tensor Flow, PyTorch, Hugging Face, Pinecone, Weights and Biases

**Programming & Systems:** Python (Pandas, NumPy, scikit-learn), C++, SQL, Linux, Git, Distributed Computing, Spark/PySpark, PostgreSQL, Redis, Airflow, Parquet, Flash, APIs

**Cloud, Infrastructure & DevOps:** AWS, GCP, Docker, Kubernetes, OpenShift, CI/CD, Grafana

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## EXPERIENCE

### UCSD [Triton Quantitative Trading Club](#) - Lead, Strategy Development Research Group

March 2026 - *present*

- **Project:** Developed delta-hedged SPY options strategies with vega-targeted sizing, IV regime filters, and automated Greeks-based execution logic; achieved Sharpe ratios of 1.3–1.7 across multi-regime backtests while reducing drawdowns through dynamic hedging. 🌐

### UCSD [Optical & Infrared Laboratory](#) - Graduate Student Researcher

April 2025 - *present*

- **Award** → [High Energy Physics Consortium for Advanced Training Fellowship](#) → *fellowship is ongoing*
- **Award** → [University of California, San Diego Sloan Scholars Fellowship](#) → *fellowship is ongoing*
- **Project:** Built QPulse, a multi-asset signal generation platform ingesting SEC filings, financial news, and market data using distributed ETL pipelines, Redis caching, and RAG-based LLM workflows on AWS/Docker/Kubernetes; processed 50k+ financial events daily while reducing retrieval latency by ~42%.
- **Project:** Constructed an in-database ML forecasting pipeline using MySQL HeatWave and vectorized query execution, reducing inference latency by 38% relative to external Python baselines across financial forecasting worklds. 🌐
- **Project:** (PhD Dissertation) Assembled and validated SiPM detector systems for nanosecond-scale optical transient detection; implemented parallelized signal-processing workflows reducing analysis runtime by ~60% across high-dim. datasets. 🌐

### UCSD Cosmology Group - Graduate Student Researcher

August 2022 - April 2025

- **Award** → [GEM Fellowship Program](#)
- **Project:** (M.S. Thesis) Studied detector non-idealities and simulated detector responses; developed stochastic signal models to quantify systematic effects on SAT-1 science products (co-authored SPIE 2024). 🌐

### NASA Goddard Space Flight Center - NASA GSFC Intern (x3)

Aug. 2021 - Aug. 2022

- **Awards** → [NASA Sally Ride Scholarship](#), [NASA Space Generation Advisory Council Scholarship](#)
- **Main Project:** Delivered single-user relay and handover experiments that demonstrated user data services for optical communications, defining scheduling parameters and spacecraft configurations to support LCRD mission operations.

### Cal Poly, Humboldt, Gravitational Physics Lab - Student Researcher

February 2019 - July 2022

- **Award** → [Cal-Bridge Program Scholarship](#)
- **Project:** (Senior Thesis) Optimized an active servo-leveling control system for a precision short-range gravity experiment using numerical modeling and sensor feedback analysis; published in J. Undergrad. Rep. Phys. (2023). 🌐

### Montana State University Solar Physics REU - Student Researcher 🌐

June - August 2021

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## EDUCATION

- University of California, San Diego | **PhD Candidate in Physics** *Expected Spring 2028*
  - University of California, San Diego | **MSc in Physics** grad. June 2025
  - California State Polytechnic University, Humboldt | **BSc in Physics, Cum Laude** grad. Dec. 2021
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## PUBLICATIONS

- **Johnson, A.** et al. *Optimization of an Active Leveling Scheme for a Short-Range Gravity Experiment.* [J. Undergrad. Rep. Phys. 33, 100002](#) (2023)
- Bhimani, Sanah, et al. (incl. **Johnson, A.**). *"The Simons Observatory: Deployment and current configuration of the Observatory Control System for SAT-MF1 and data access software systems."* [Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII. Vol. 13102](#). SPIE, 2024.